

4				3				2				1						
BOM TABLE																		
ITEM NO.		PART NUMBER			DESCRIPTION			QTY.	MATERIAL PROPERTY			UNIT		TOTAL COST				
1		RR602-11			CHASIS			1	2			FREE		FREE				
2		RR602-12			WHEELHOLD(1-2)			2	2			FREE		FREE				
3		RR602-13			9V BATTERY HOLDER			1	15			FREE		FREE				
4		RR602-14			6AA HOLDER			1	15			0.4		0.4				
5		RR602-15			BUCKET_DROP			1	2			FREE		FREE				
6		RR602-16			BUCKET_BASE			1	2			FREE		FREEE				
7		RR602-17			MOTOR_MOUNT			1	2			FREE		FREE				
8		QTR-8A			LINE SENSOR			1	1			FREE		FREE				
9		ARDUINO UNO R3_PIN.STP			MICRO CONTROLLER			1	1			FREE		FREE				
10		L298N			MOTOR DRIVER			1	1			FREE		FREE				
11		MOTORDCTT			BLUE TT MOTOR			1	1			FREE		FREE				
12		MOTORDCTTPIEZA			MOTOR SHAFT			1	1			FREE		FREE				
13		X187 OLDGRAY TECHNIC GEAR 24 TOOTH WITH THREE AXLEHOLES			24 TOOTH GEAR			4	14			0.135		0.54				
14		RR602-18			LINKAGE			1	2			FREE		FREE				
15		LARGE_WHEEL68.8X36MM			DRIVER WHEEL			1	14			7		7				
16		4019 OLDGRAY TECHNIC GEAR 16 TOOTH			16 TOOTH GEAR			3	14			0.135		0.27				
17		LEGOAXEL39.5mm			INTERMEDIATE AXEL			4	14			FREE		FREE				
18		LEGOAXEL95.3mm			DRIVER AXEL			1	14			FREE		FREE				
19		STAINLESSSTEELBEARING0.86IN			LARGE BEARING			2	14			0.5		1				
20		STAINLESSSTEELBEARING0.45IN			SMALL BEARING			2	14			0.5		1				
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									DIMENSIONS ARE IN INCHES		DRAWN		VAUGHN					4/20/2026
									TOLERANCES:		CHECKED		JACK					4/20/2026
									FRACTIONAL ±		ENG APPR.							
									ANGULAR: MACH ± BEND ±		MFG APPR.							
									TWO PLACE DECIMAL ±		Q.A.							
									THREE PLACE DECIMAL ±		COMMENTS:							
									INTERPRET GEOMETRIC TOLERANCING PER:									
									MATERIAL					SIZE	DWG. NO.		REV	
					NEXT ASSY		USED ON		FINISH					B	RR602-01		1	
					APPLICATION			DO NOT SCALE DRAWING						SCALE: 1:4	WEIGHT: 1.5		SHEET 1 OF 20	
4				3				2				1						
SOLIDWORKS Educational Product. For Instructional Use Only.																		

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BOM TABLE															
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.	MATERIAL PROPERTY	UNIT	TOTAL COST									
21	RR602-19	SMALL BEARING ADAPTER	2	15	.009	0.018									
22	RR602-20	LARGE BEARING ADAPTER	2	15	.04	0.08									
23	LEGO_GEAR_40_TOOTH	40 TOOTH GEAR	2	14	0.135	0.27									
24	RR602-21	MOTOR TO GEAR ADAPTER	2	15	0.07	0.14									
25	LEGOAXEL16MM	MOTOR TO GEAR AXEL	3	14	FREE	FREE									
26	9G SERVO MOTOR SG90	SERVO	1	1	FREE	FREE									
27	RR602-22	SERVO HOLDER	1	15	.07	0.07									
28	11 DELIK LIFTARM	PITMAN ARM	1	14	FREE	FREE									
29	LEGO TECHNIC AXEL PIN CONECT	LARGE LEGO SPACER	2	14	FREE	FREE									
30	LEGO4X2STEERARM	L LEGO 4X2	2	14	FREE	FREE									
31	LEGOTECHNIC90	90 DEG LEGO	1	14	FREE	FREE									
32	LEGOTECHNICSPACER	SMALL LEGO SPACER	6	14	FREE	FREE									
33	LEGOTECHNICSTEERINGCONNECTOR	WHEEL TO PITMAN LEGO	2	14	FREE	FREE									
34	LEGOTECHNICTHRUPIN	LEGO THRU PIN	2	14	FREE	FREE									
35	MG90S_SERVOHORN	SERVO HORN	1	1	FREE	FREE									
36	SMALL_TIRE43.2X22MM	FRONT WHEEL	2	14	0.33	0.66									
37	LEGOTECHNICAXEL32MM	STEERING AXEL	2	14	FREE	FREE									
38	B18.6.7M - M3 X 0.5 X 13 TYPE I CROSS RECESSED PHMS --13N	FRAME SCREWS	8	10	FREE	FREE									
39	B18.6.7M - M3 X 0.5 X 4 TYPE I CROSS RECESSED PHMS --4N	ELECTRICAL SCREWS	4	10	FREE	FREE									
40	B18.6.7M - M2 X 0.4 X 10 TYPE I CROSS RECESSED PHMS --10N	MOTOR MOUNT SCREWS	4	10	FREE	FREE									
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					DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ± THREE PLACE DECIMAL ±	DRAWN	VAUGHN	4/20/2026							
						CHECKED	JACK	4/20/2026							
						ENG APPR.									
						MFG APPR.									
					INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.			COMMENTS:						
					MATERIAL										
NEXT ASSY	USED ON	FINISH				SIZE	DWG. NO.	REV							
APPLICATION		DO NOT SCALE DRAWING				B	RR602-01	1							
						SCALE: 1:4		WEIGHT: 1.5		SHEET 2 OF 20					

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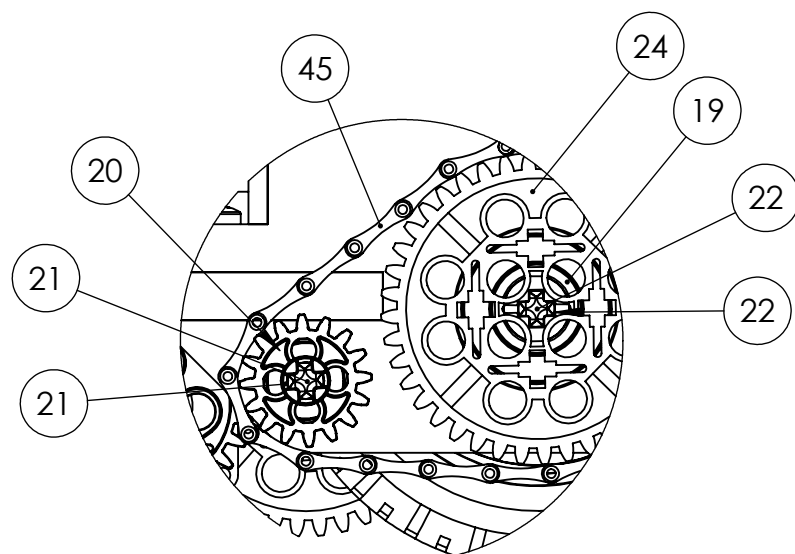
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ITEM NO.	PART NUMBER	DESCRIPTION	QTY.	MATERIAL PROPERTY	COST	TOTAL COST																																																													
41	B18.6.7M - M2 X 0.4 X 8 TYPE I CROSS RECESSED PHMS --8N	SERVO SCREW	1	10	FREE	FREE																																																													
42	B18.6.7M - M2.5 X 0.45 X 3 TYPE I CROSS RECESSED PHMS --3N	SERVO HORN SCREW	1	10	FREE	FREE																																																													
43	B18.6.7M - M2 X 0.4 X 2.5 TYPE I CROSS RECESSED PHMS --2.5N	LINE SENSOR SCREWS	2	10	FREE	FREE																																																													
44	COVER	ROBOT COVER	1	15	FREE	FREE																																																													
45	LINKS	CHAIN LINKS	20	14	FREE	FREE																																																													
46	NEO_PIXEL	STATE LED	1	16	FREE	0																																																													
47	PROTO_BOARD	PROTOBOARD	1	16	FREE	0																																																													
48	PUSH_BUTTON	PUSHBUTTON	2	16	FREE	0																																																													
49	RR602-02	CHASSIS ASSEMBLY	1	2	FREE	FREE																																																													
50	RR602-03	WHEEL ASSEMBLY	2	14	FREE	FREE																																																													
51	RR602-04	COVER ASSEMBLY	1	2	FREE	FREE																																																													
52	RHFNBOLT 0.19-24x1.25x.75N	HITCH BOLT	2	10	FREE	FREE																																																													
53	HNUT 0.25000-20-D-N	HITCH NUT	3	10	FREE	FREE																																																													
54	FW 0.203	HITCH WASHER	2	10	FREE	FREE																																																													
-	-	TOTAL COST	-	-	-	11.60																																																													
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APPLICATION		DO NOT SCALE DRAWING		SCALE: 1:4 WEIGHT: 1.5 SHEET 3 OF 20																																																															

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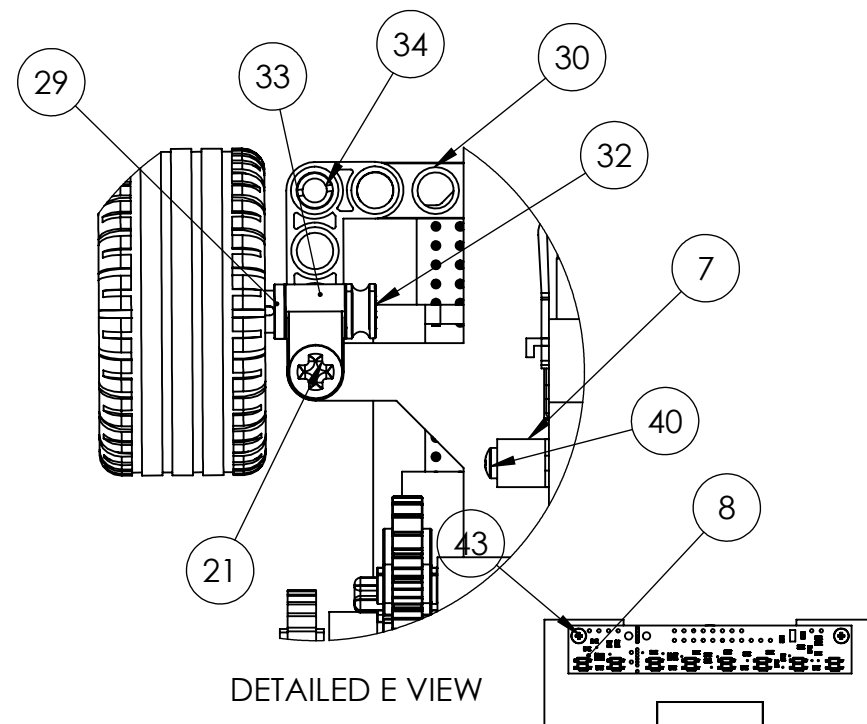
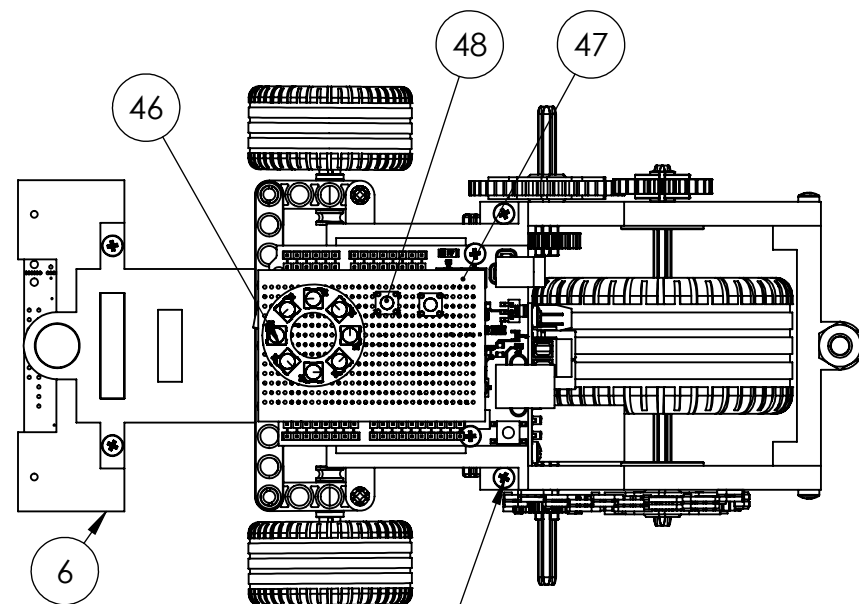
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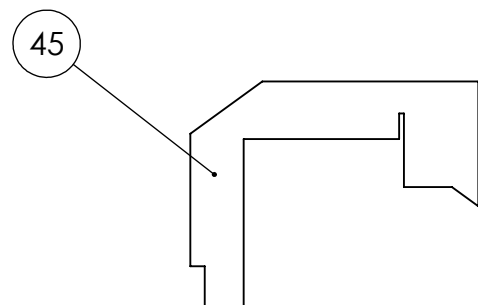
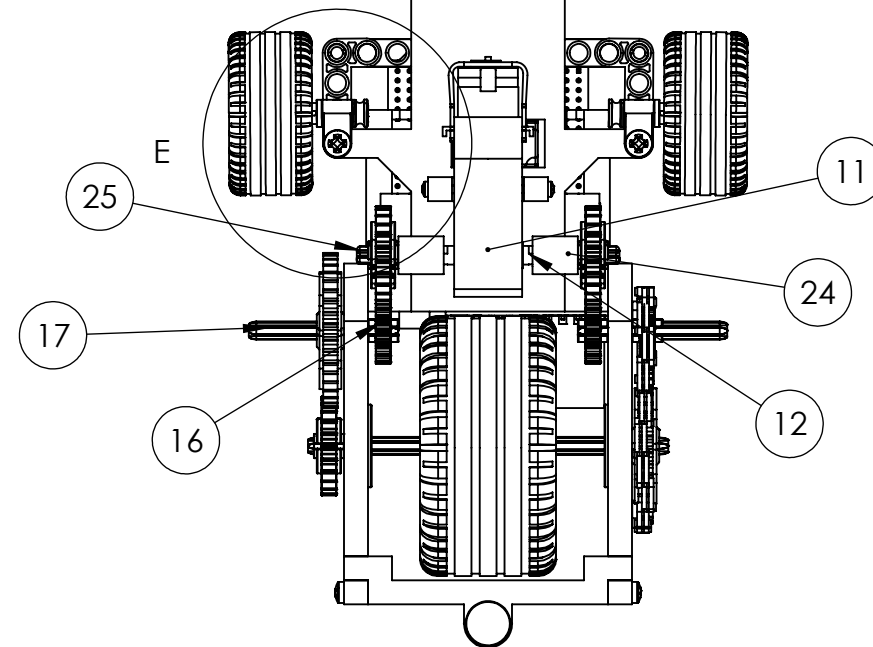
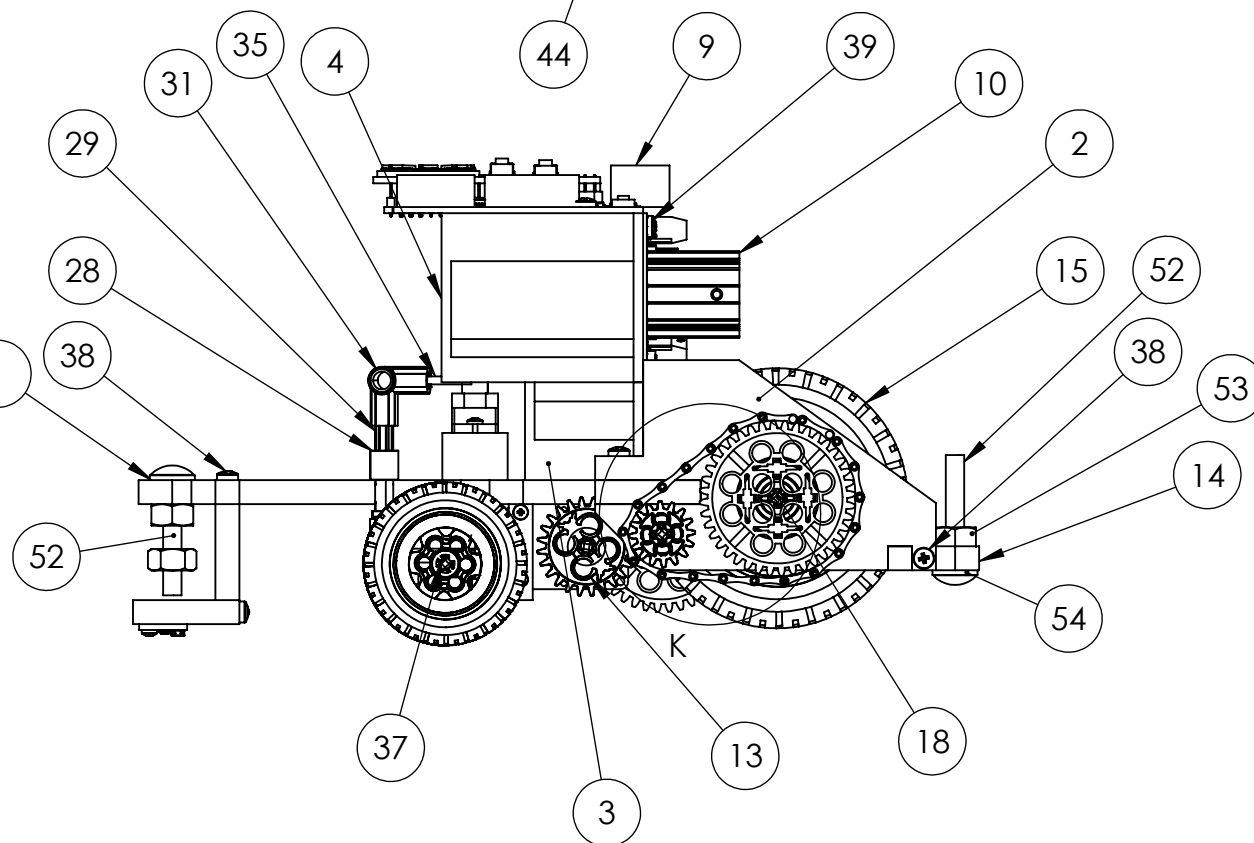
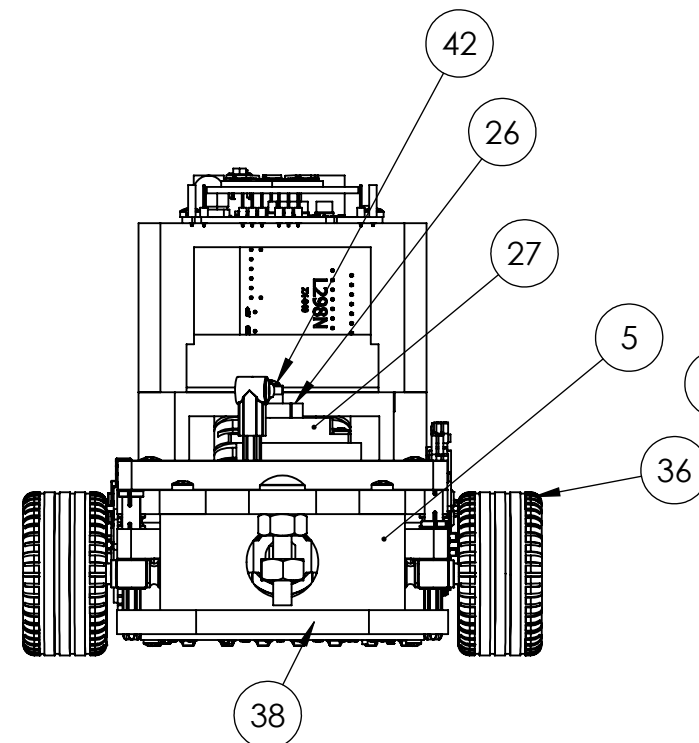
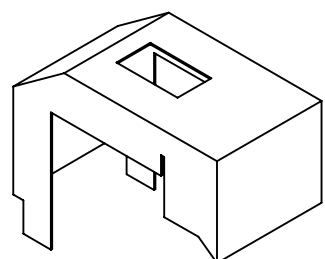
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DETAILED K VIEW



DETAILED E VIEW

COVER REMOVED FOR VISUAL PURPOSES
SCALE: 1:4

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DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL $\pm$
ANGULAR: MACH $\pm$ BEND $\pm$
TWO PLACE DECIMAL $\pm$
THREE PLACE DECIMAL $\pm$

INTERPRET GEOMETRIC
TOLERANCING PER:

MATERIAL

FINISH

NEXT ASSY

USED ON

APPLICATION

DO NOT SCALE DRAWING

NAME DATE

DRAWN VAUGHN 4/20/2026

CHECKED JACK 4/20/2026

ENG APPR.

MFG APPR.

Q.A.

COMMENTS:

TITLE:

ROBO RACER

SIZE DWG. NO.

B RR602-01

REV

1

SCALE: 1:2 WEIGHT: 1.5

SHEET 4 OF 20

3

2

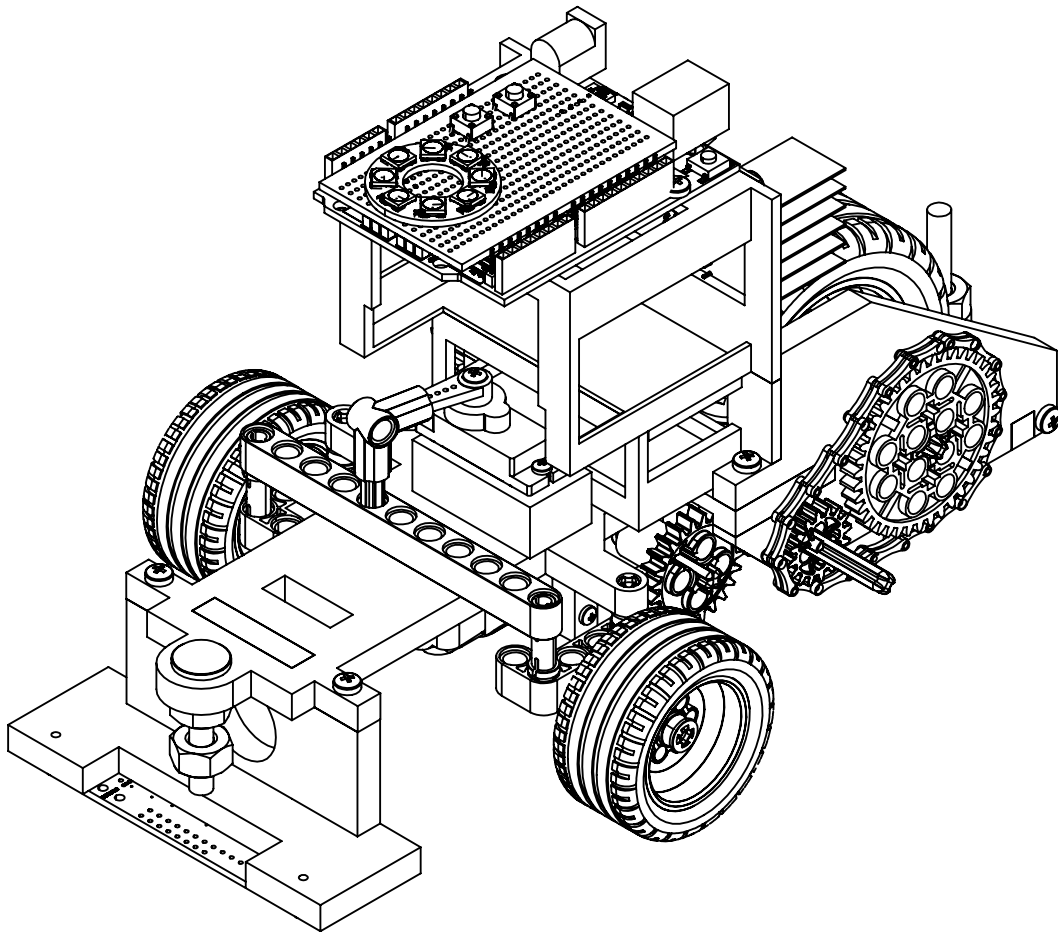
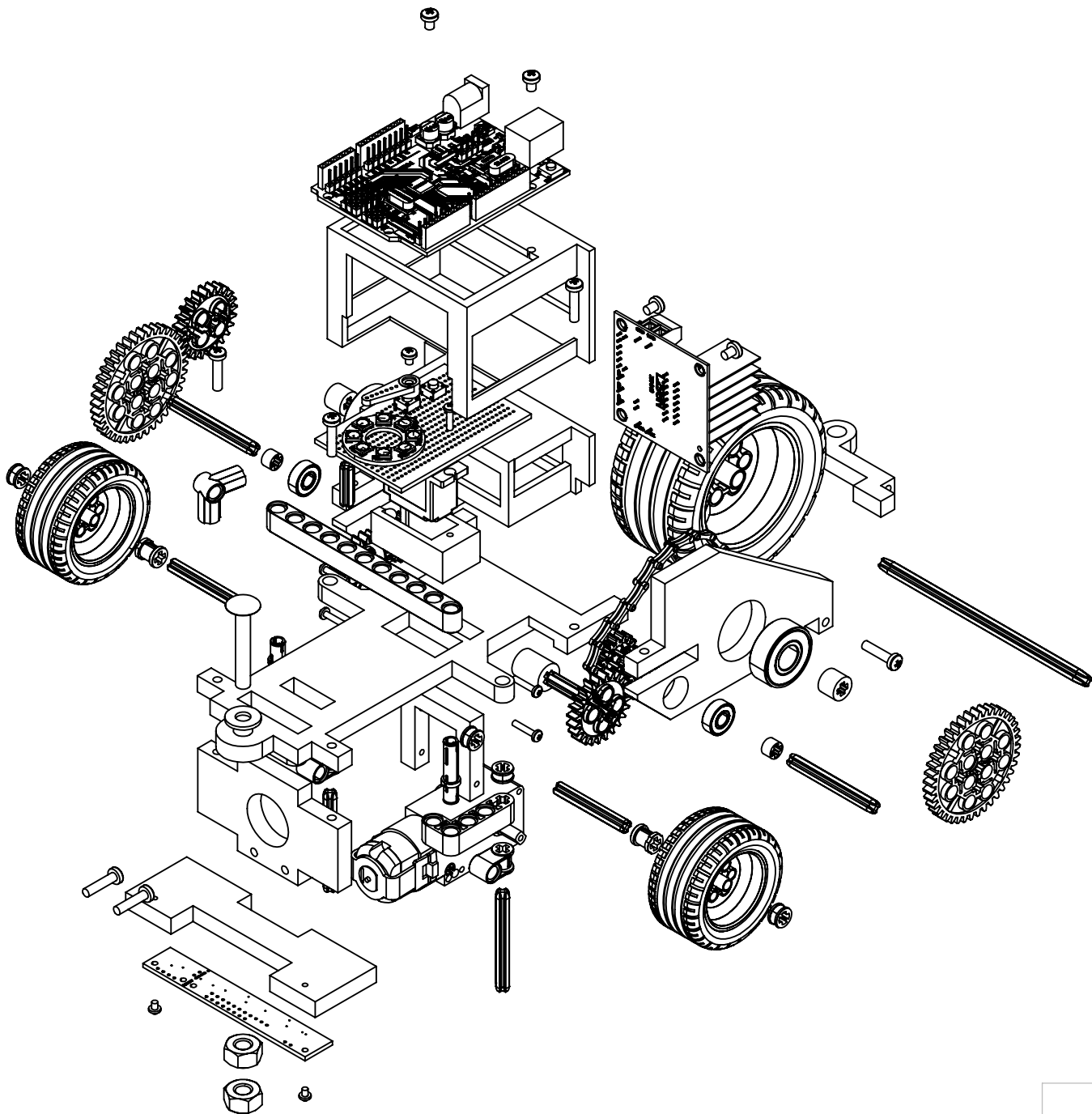
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		FRACTIONAL ±	ENG APPR.				
		ANGULAR: MACH ± BEND ±	MFG APPR.				
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		THREE PLACE DECIMAL ±	COMMENTS:				
		INTERPRET GEOMETRIC TOLERANCING PER:					
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NEXT ASSY	USED ON	FINISH				B	RR602-01
APPLICATION		DO NOT SCALE DRAWING					1

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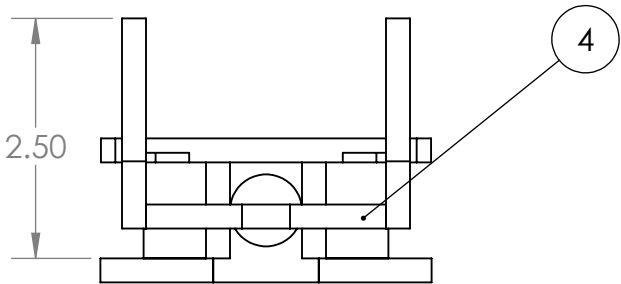
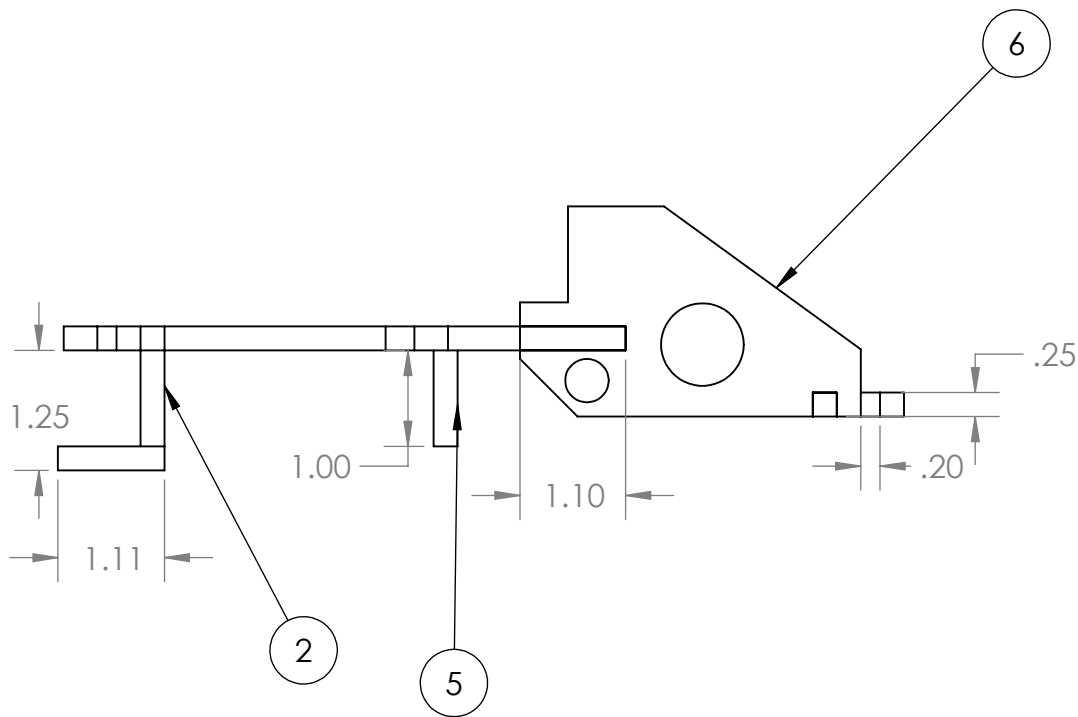
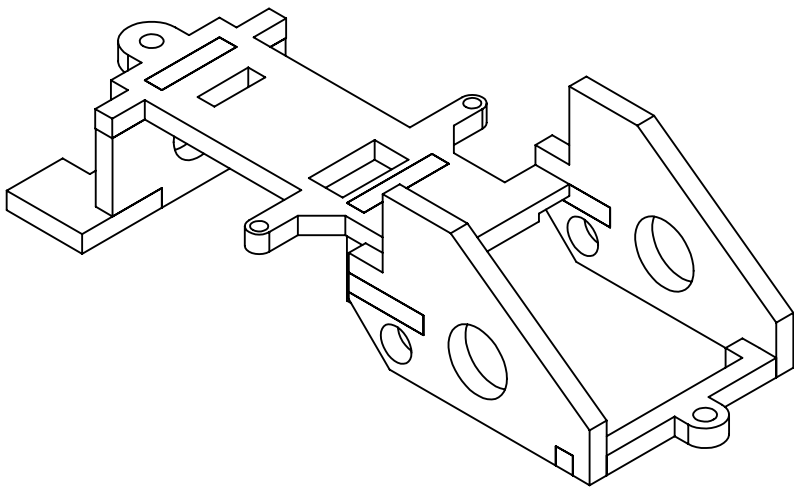
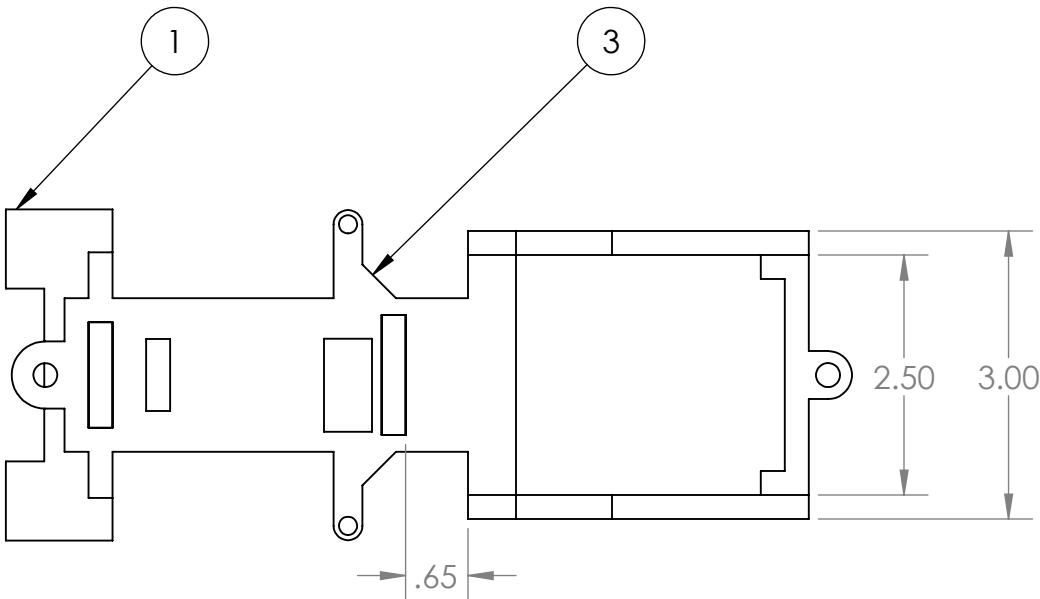
4

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1

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	RR602-16	BUCKET_BASE	1
2	RR602-15	BUCKET_DROP	1
3	RR602-11	CHASIS	1
4	RR602-18	LINKAGE	1
5	RR602-17	MOTOR_MOUNT	1
6	RR602-12	WHEELHOLD(1-2)	2



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			ENG APPR.				
		INTERPRET GEOMETRIC TOLERANCING PER:	MFG APPR.			SIZE DWG. NO. REV B RR602-02 1	
		MATERIAL NYLON	Q.A.				
NEXT ASSY	USED ON	FINISH	COMMENTS:			SCALE: 1:2	SHEET 6 OF 20
APPLICATION		DO NOT SCALE DRAWING					

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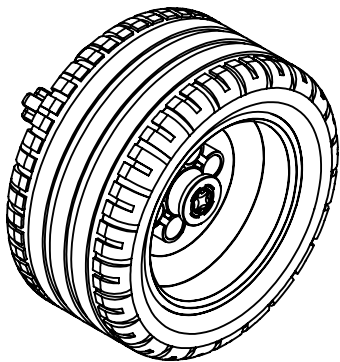
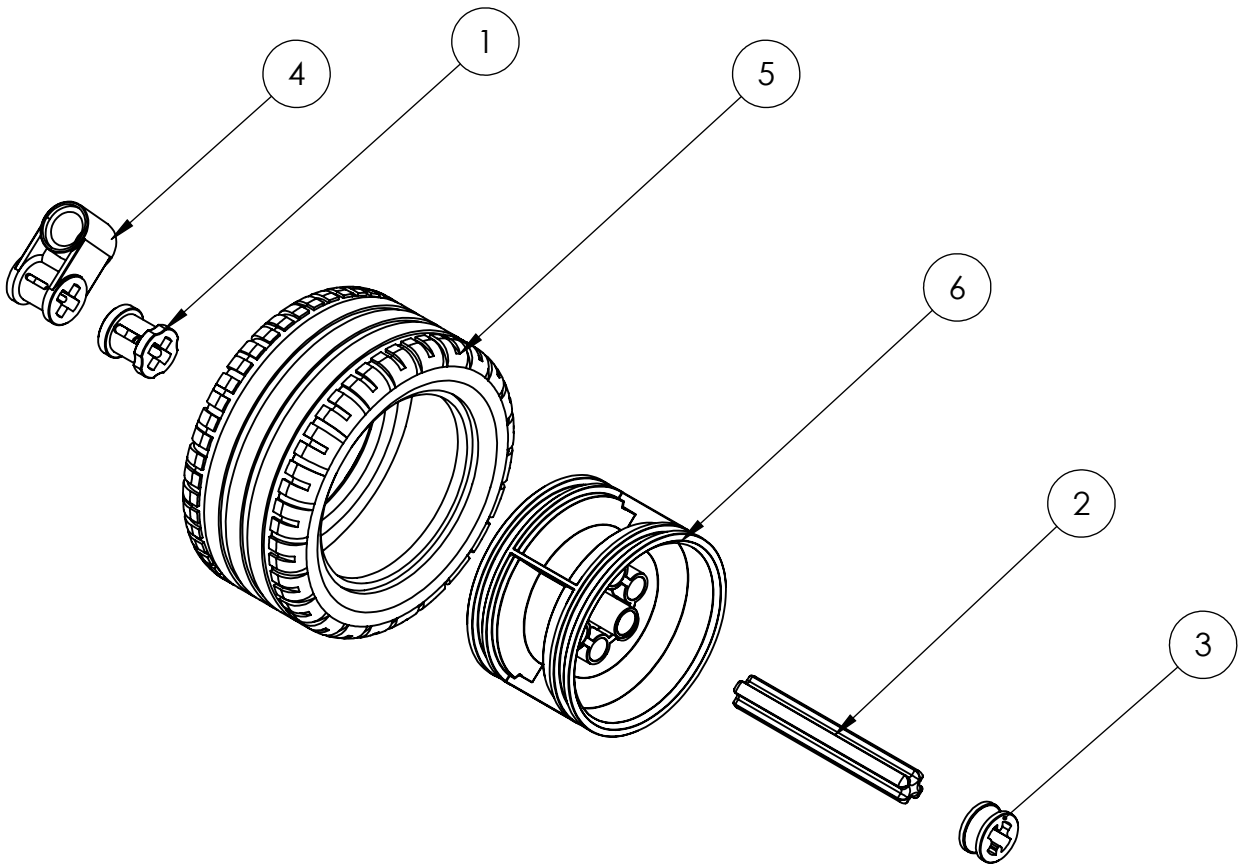
B

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ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	LEGO TECHNIC AXEL PIN CONECT	LARGE LEGO SPACER	1
2	LEGOTECHNICAXEL32 MM	STEERING AXEL	1
3	LEGOTECHNICSPACER	SMALL LEGO SPACER	1
4	LEGOTECHNICSTEERIN GCONNECTOR	WHEEL TO PITMAN LEGO	1
5	SMALL_TIRE	LEGO STEER TIRE	1
6	SMALL_TIRE_RIM	LEGO TIRE RIM	1



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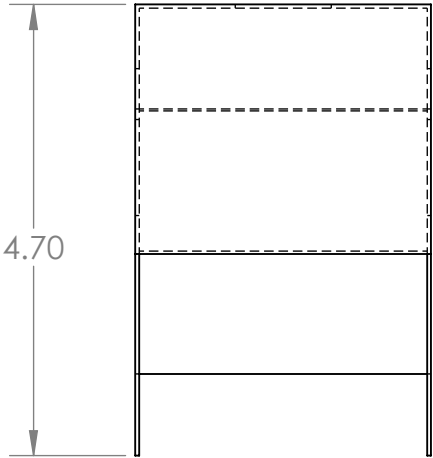
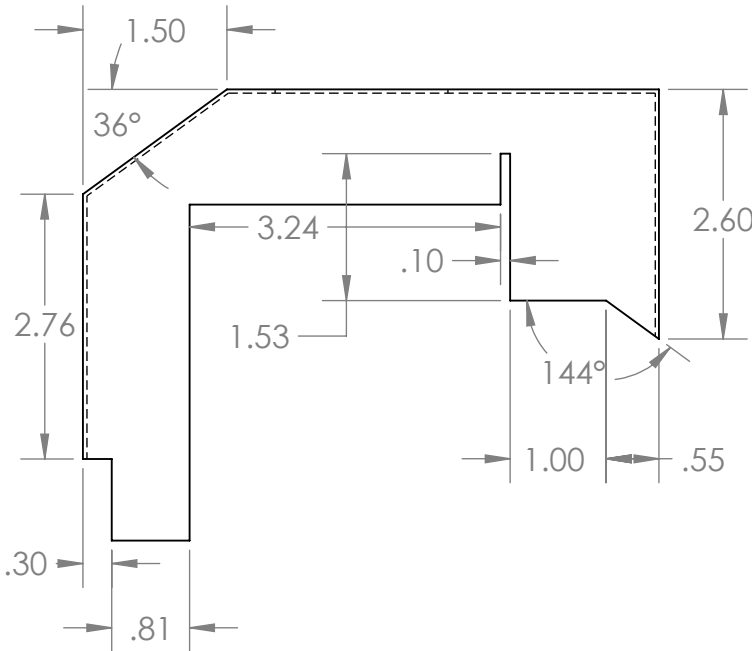
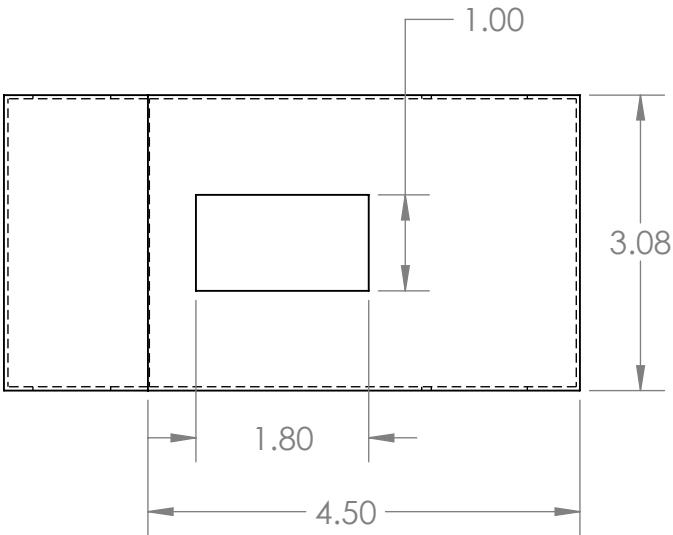
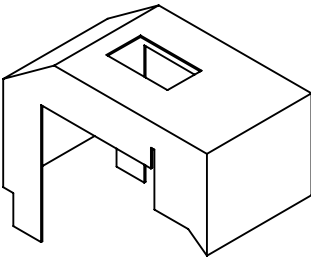
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		ANGULAR: MACH ± BEND ±	ENG APPR.					
		TWO PLACE DECIMAL ±	MFG APPR.					
		THREE PLACE DECIMAL ±	Q.A.			SIZE DWG. NO. REV B RR602-03 1		
		INTERPRET GEOMETRIC TOLERANCING PER:	COMMENTS:					
		MATERIAL LEGO						
		FINISH						
NEXT ASSY	USED ON							
APPLICATION		DO NOT SCALE DRAWING						
			SCALE: 1:1			SHEET 7 OF 20		

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THICKNESS OF ENTIRE PART IS 0.04

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		TOLERANCES: FRACTIONAL ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ± THREE PLACE DECIMAL ±	CHECKED	JACK	4/20/2026			
			ENG APPR.					
		INTERPRET GEOMETRIC TOLERANCING PER:	MFG APPR.			SIZE DWG. NO. REV B RR602-04 1		
		MATERIAL PLA	Q.A.					
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NEXT ASSY	USED ON					SCALE: 1:2		SHEET 8 OF 20
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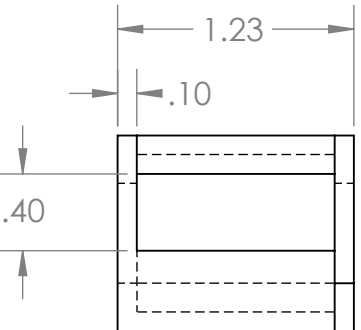
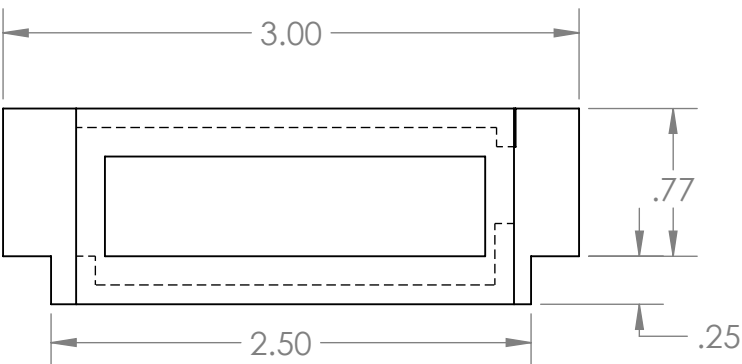
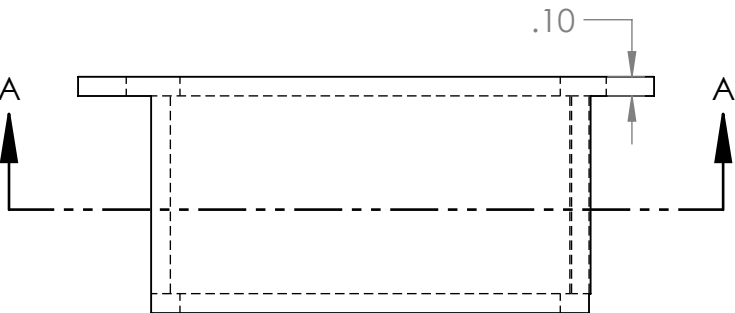
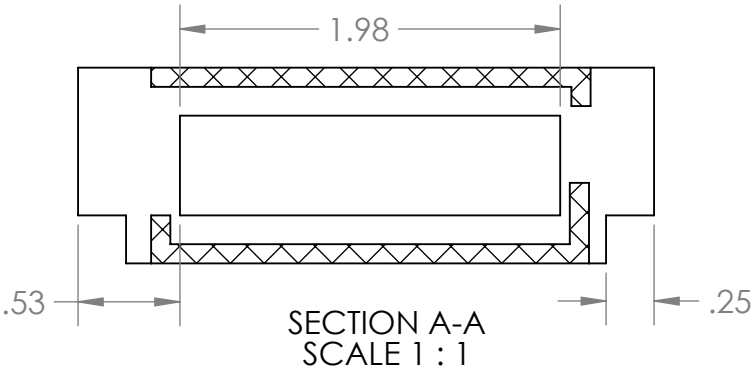
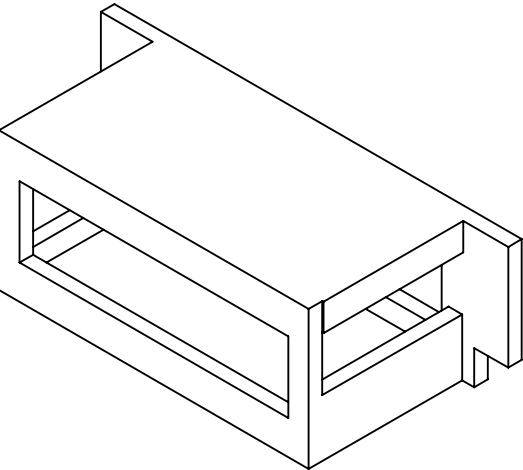
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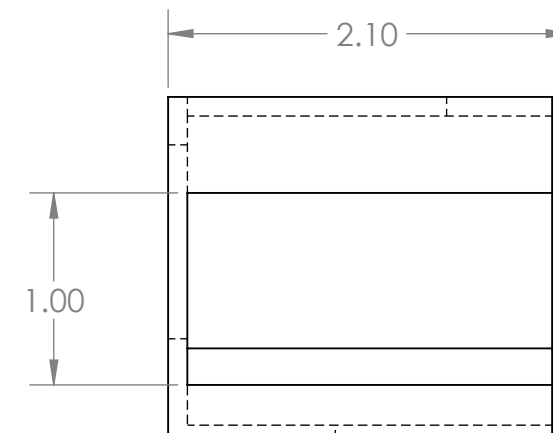
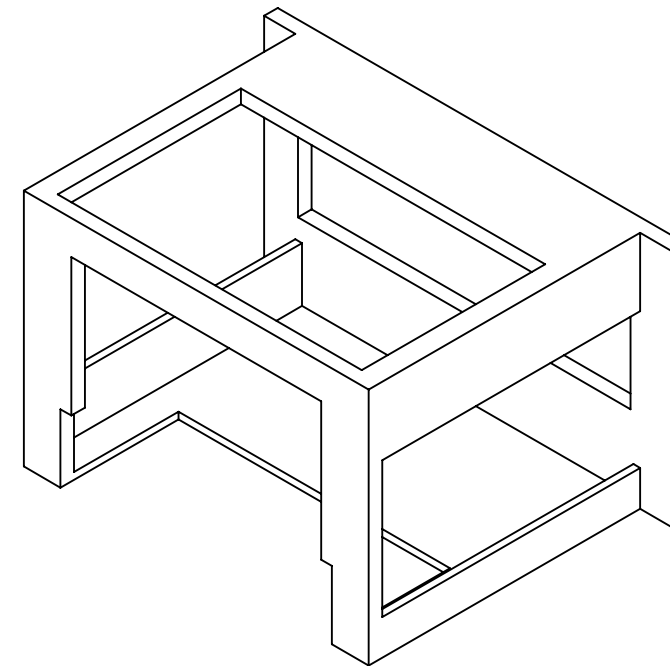
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		TWO PLACE DECIMAL ±	Q.A.			SIZE	DWG. NO.	REV
		THREE PLACE DECIMAL ±	COMMENTS:			B	RR602-13	1
NEXT ASSY	USED ON	MATERIAL PLA				SCALE: 1:1		SHEET 9 OF 20
		FINISH						
		DO NOT SCALE DRAWING						

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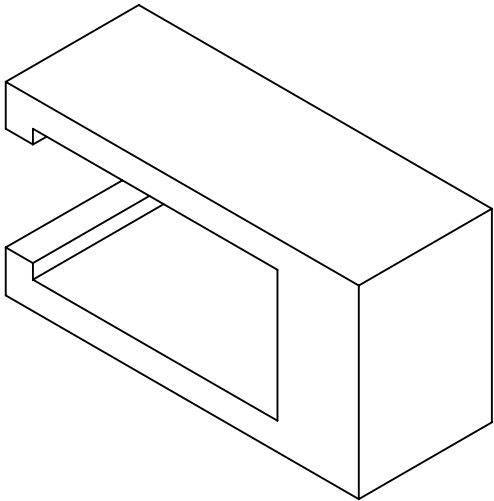
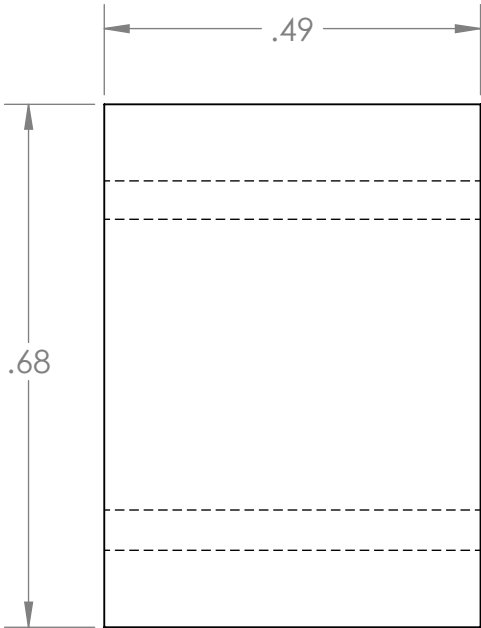
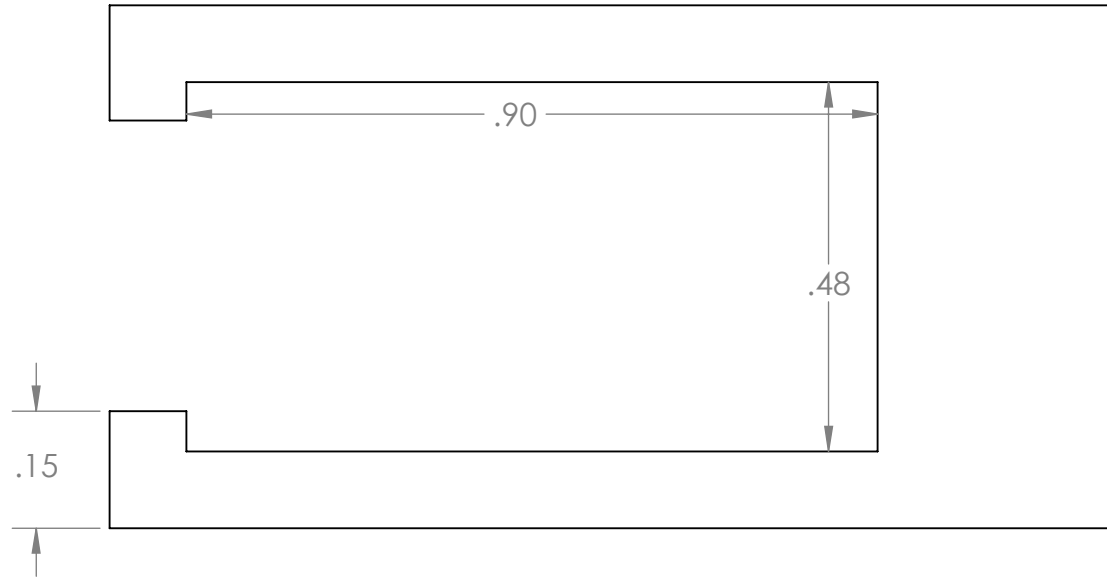
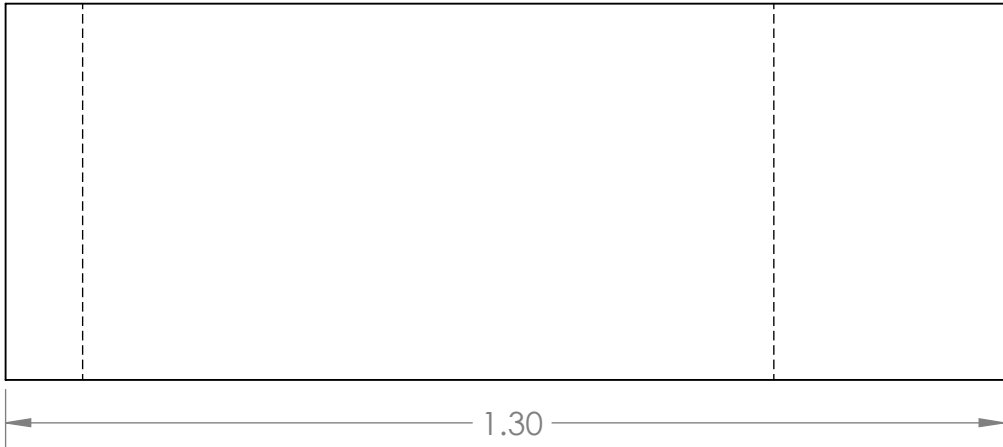
SHEET 10 OF 20

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		ANGULAR: MACH ± BEND ±	ENG APPR.					
		TWO PLACE DECIMAL ±	MFG APPR.					
		THREE PLACE DECIMAL ±	Q.A.			SIZE DWG. NO. REV B RR602-22 1		
		INTERPRET GEOMETRIC TOLERANCING PER:	COMMENTS:					
		MATERIAL PLA						
		FINISH						
NEXT ASSY	USED ON							
APPLICATION		DO NOT SCALE DRAWING	SCALE: 4:1			WEIGHT: 1.5	SHEET 11 OF 20	

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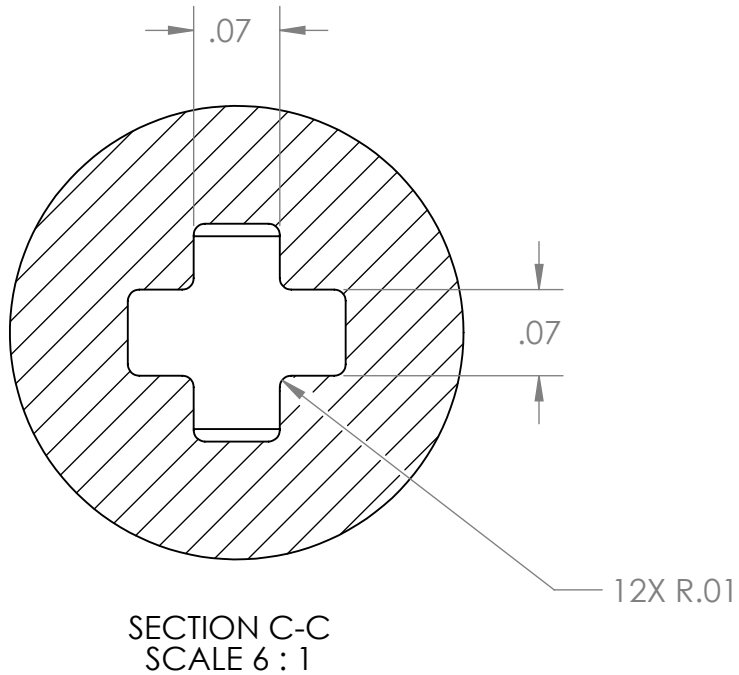
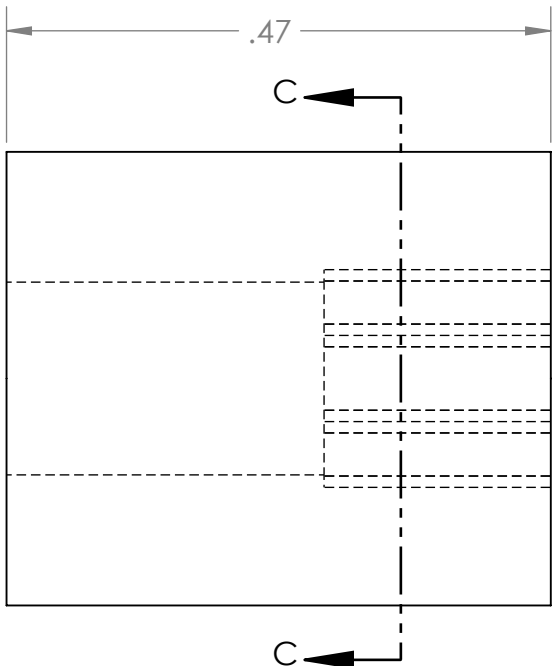
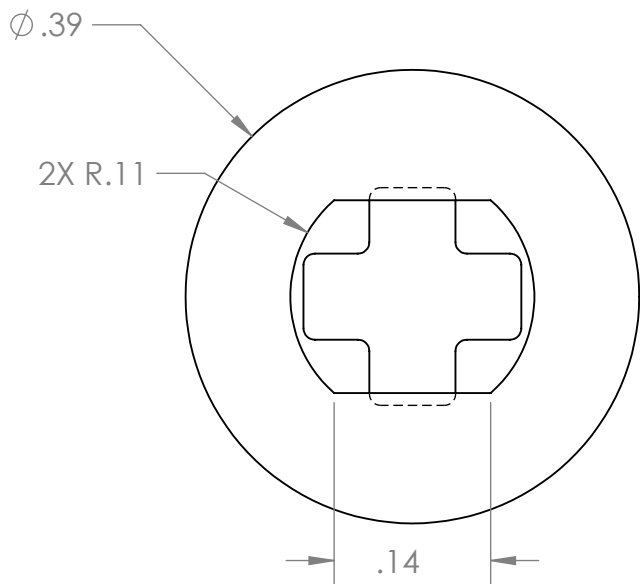
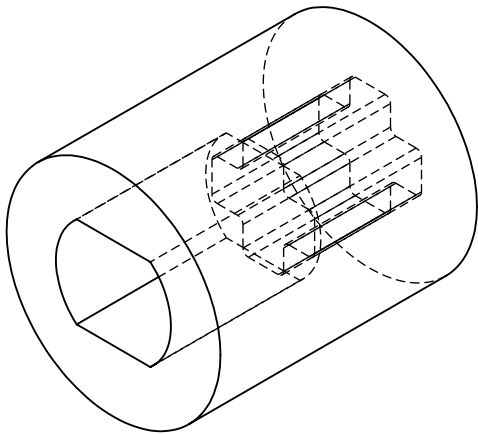
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		TOLERANCES: FRACTIONAL ±	CHECKED	JACK	4/20/2026			
		ANGULAR: MACH ± BEND ±	ENG APPR.					
		TWO PLACE DECIMAL ±	MFG APPR.					
		THREE PLACE DECIMAL ±	Q.A.			SIZE DWG. NO. REV B RR602-21 1		
		INTERPRET GEOMETRIC TOLERANCING PER:	COMMENTS:					
		MATERIAL PLA						
NEXT ASSY	USED ON	FINISH						
APPLICATION		DO NOT SCALE DRAWING				SCALE: 6:1		SHEET 12 OF 20

4

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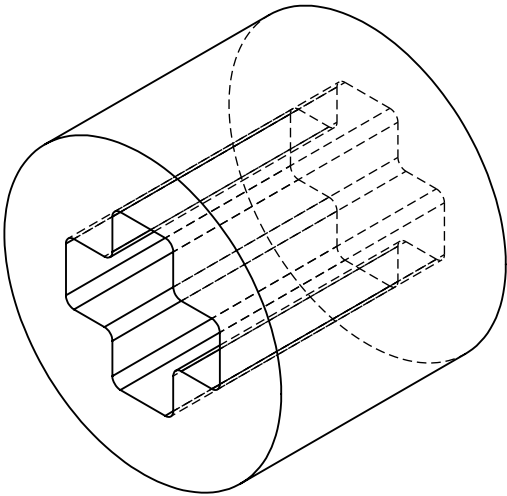
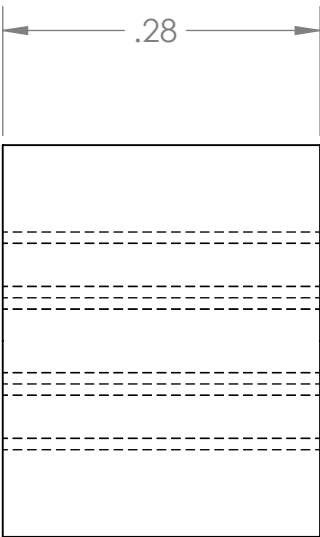
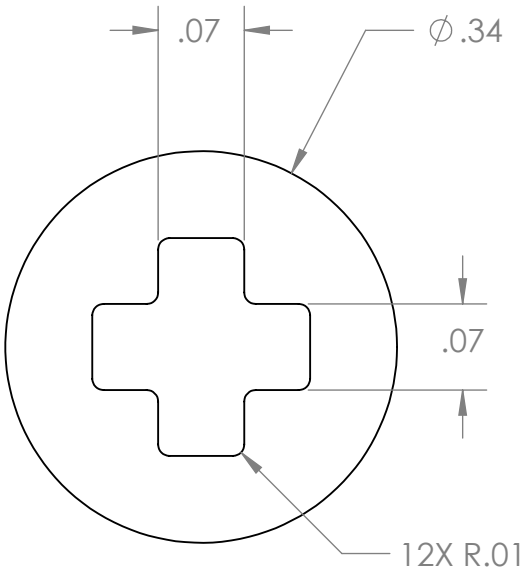
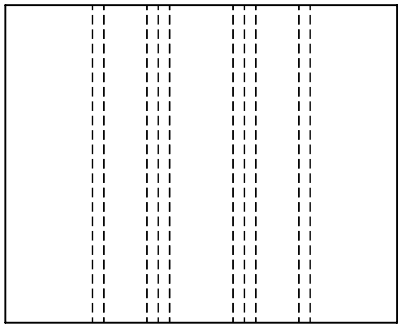
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		TOLERANCES:	CHECKED	JACK	4/20/2026			
		FRACTIONAL ±	ENG APPR.					
		ANGULAR: MACH ± BEND ±	MFG APPR.					
		TWO PLACE DECIMAL ±				SIZE DWG. NO. REV B RR602-20 1		
		THREE PLACE DECIMAL ±						
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.					
		MATERIAL	COMMENTS:			SCALE: 6:1 SHEET 13 OF 20		
		PLA						
		FINISH						
NEXT ASSY	USED ON							
APPLICATION		DO NOT SCALE DRAWING						

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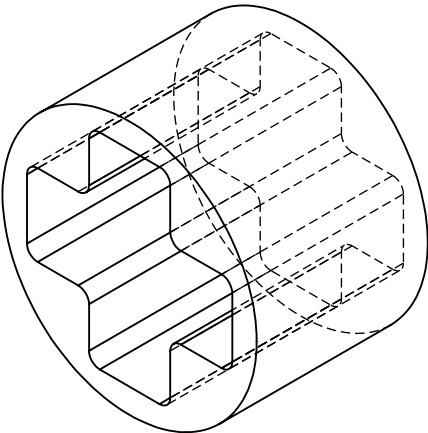
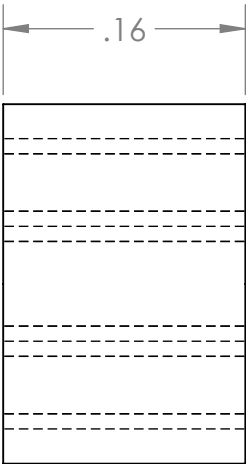
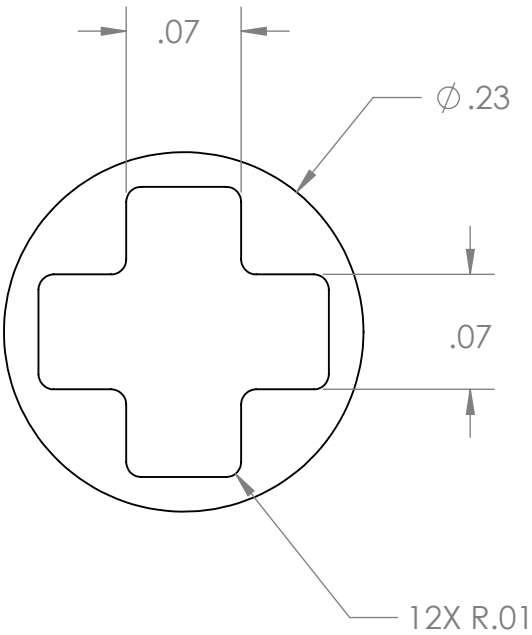
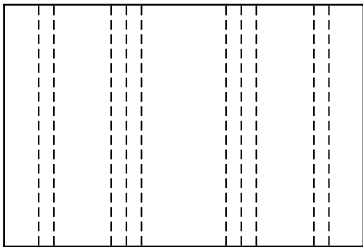
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		TOLERANCES:	CHECKED	JACK	4/20/2026			
		FRACTIONAL ±	ENG APPR.					
		ANGULAR: MACH ± BEND ±	MFG APPR.					
		TWO PLACE DECIMAL ±	Q.A.			SIZE DWG. NO. REV B RR602-19 1		
		THREE PLACE DECIMAL ±	COMMENTS:					
		INTERPRET GEOMETRIC TOLERANCING PER:						
		MATERIAL						
		STEEL						
NEXT ASSY	USED ON	FINISH						
APPLICATION		DO NOT SCALE DRAWING				SCALE: 8:1		SHEET 14 OF 20

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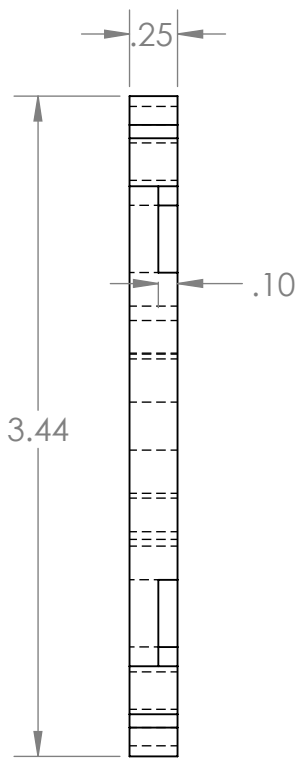
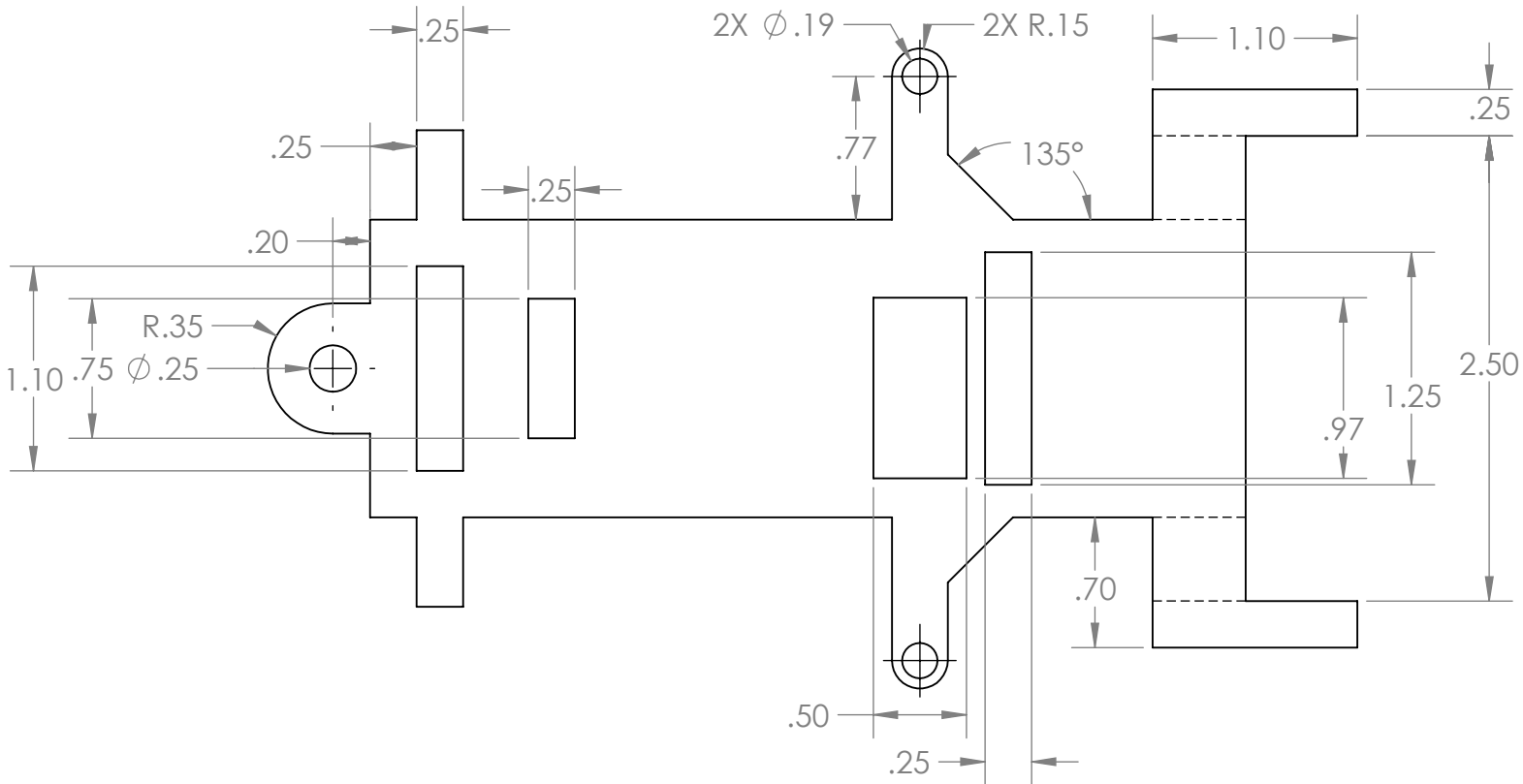
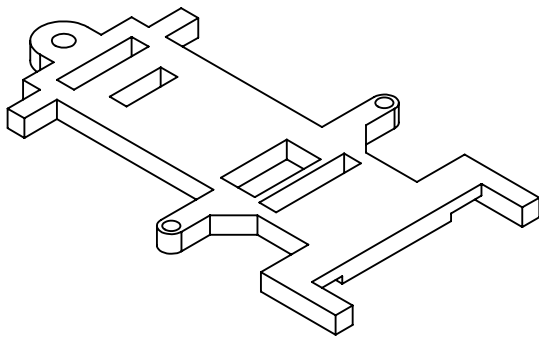
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		TWO PLACE DECIMAL ± THREE PLACE DECIMAL ±	MFG APPR.					
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.			SIZE DWG. NO. REV B RR602-11 1		
		MATERIAL NYLON	COMMENTS:					
		FINISH						
NEXT ASSY	USED ON							
APPLICATION		DO NOT SCALE DRAWING						
			SCALE: 1:1					

TITLE:

ROBO RACER

SIZE

B

DWG. NO.

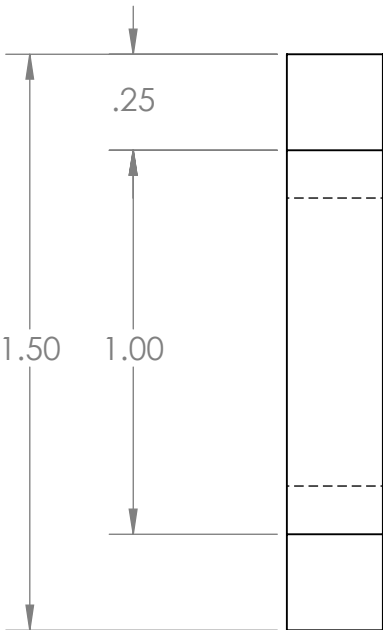
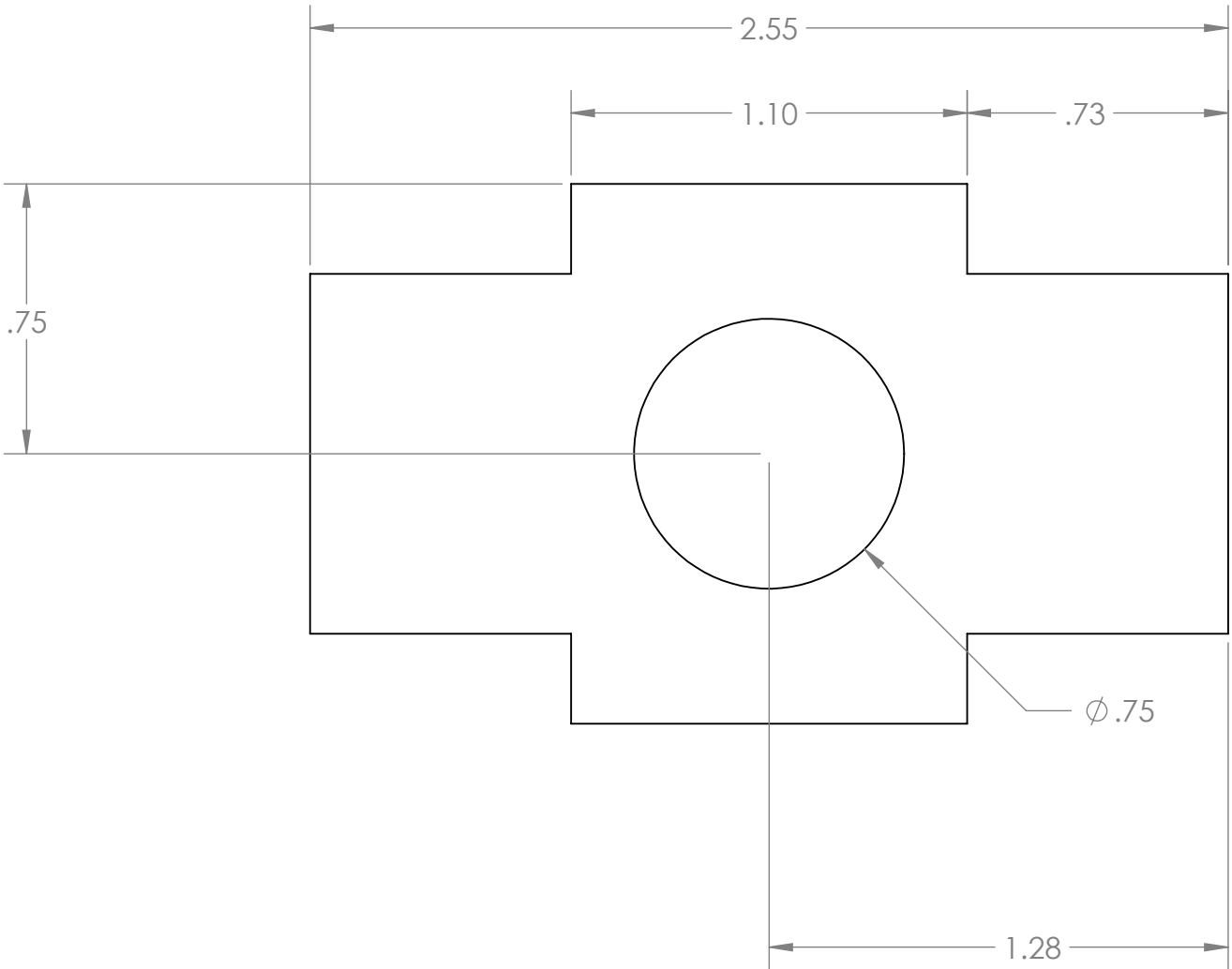
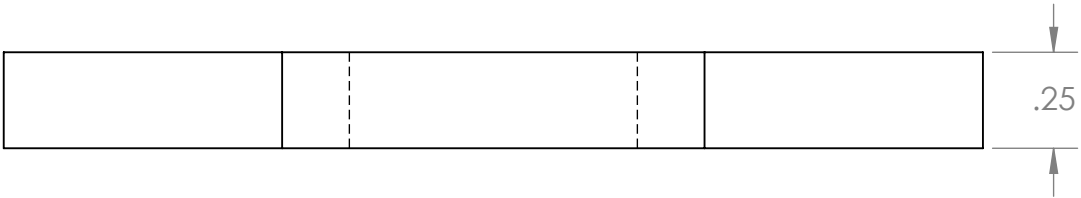
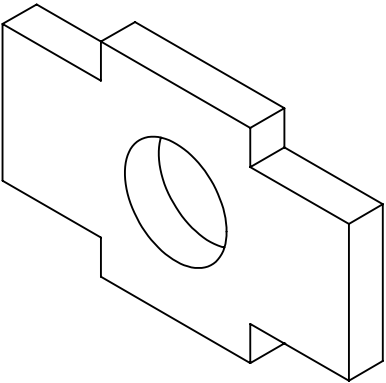
RR602-11

REV

1

SCALE: 1:1

SHEET 15 OF 20

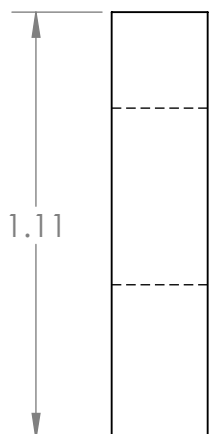
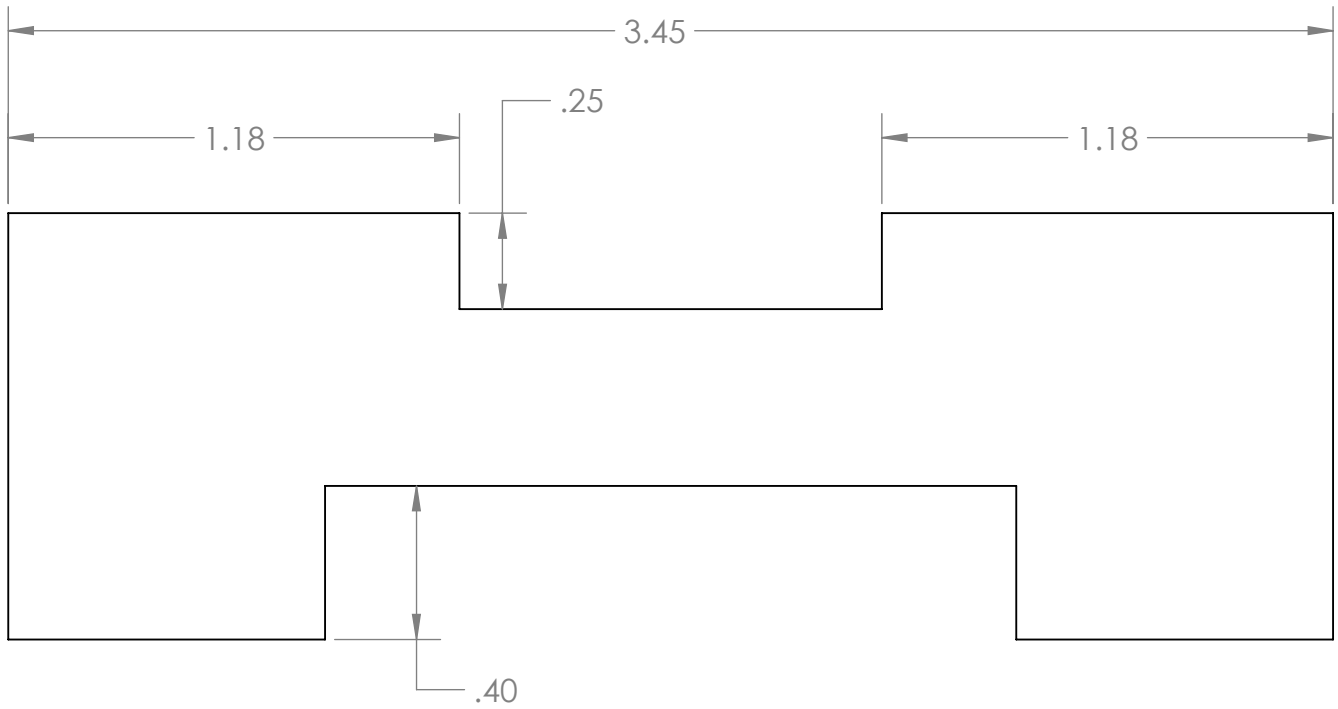
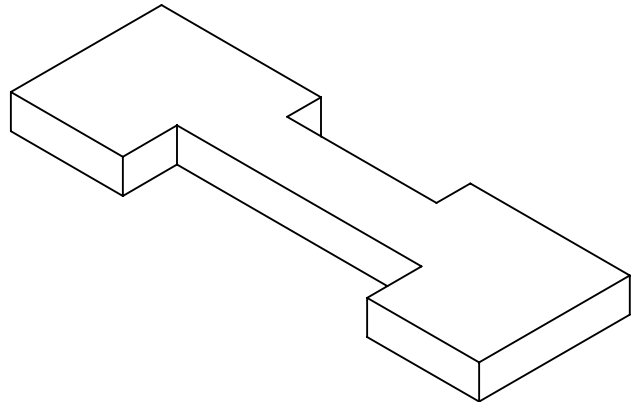
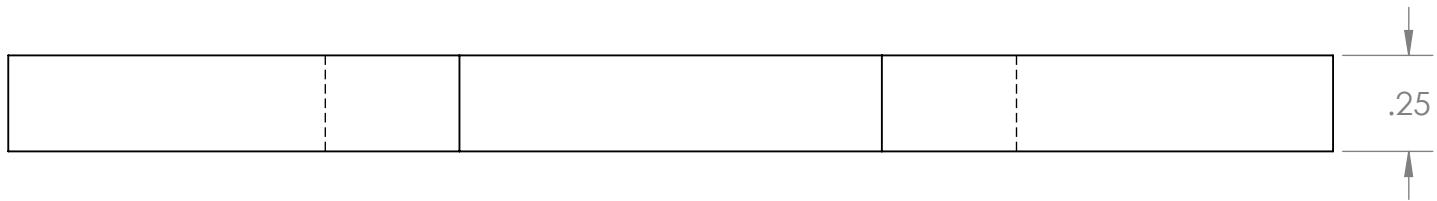


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			CHECKED	JACK	4/20/2026			
			ENG APPR.					
		INTERPRET GEOMETRIC TOLERANCING PER:				SIZE DWG. NO. REV		
		MATERIAL NYLON						
		FINISH				SCALE: 2:1 SHEET 16 OF 20		
NEXT ASSY	USED ON							
APPLICATION		DO NOT SCALE DRAWING						

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			CHECKED	JACK				4/20/2026
			ENG APPR.					
			MFG APPR.					
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.			SIZE DWG. NO. REV B RR602-16 1		
		MATERIAL	COMMENTS:					
		NYLON						
NEXT ASSY	USED ON	FINISH						
APPLICATION		DO NOT SCALE DRAWING						
					SCALE: 2:1			SHEET 17 OF 20

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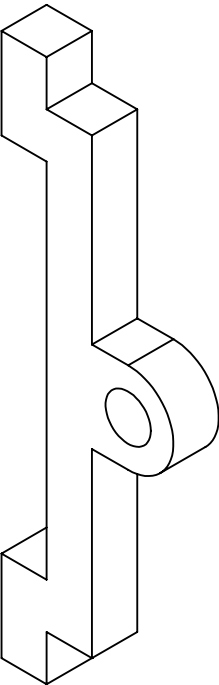
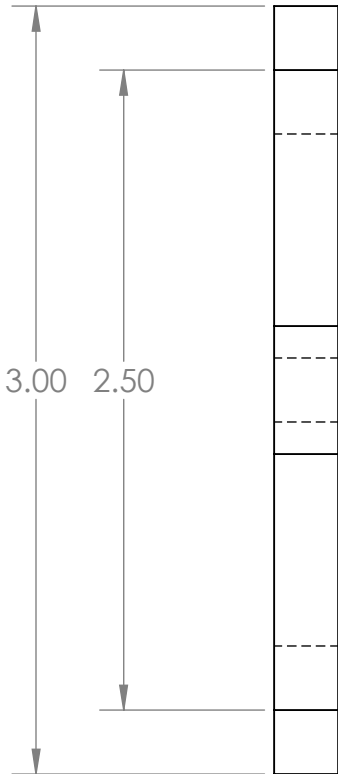
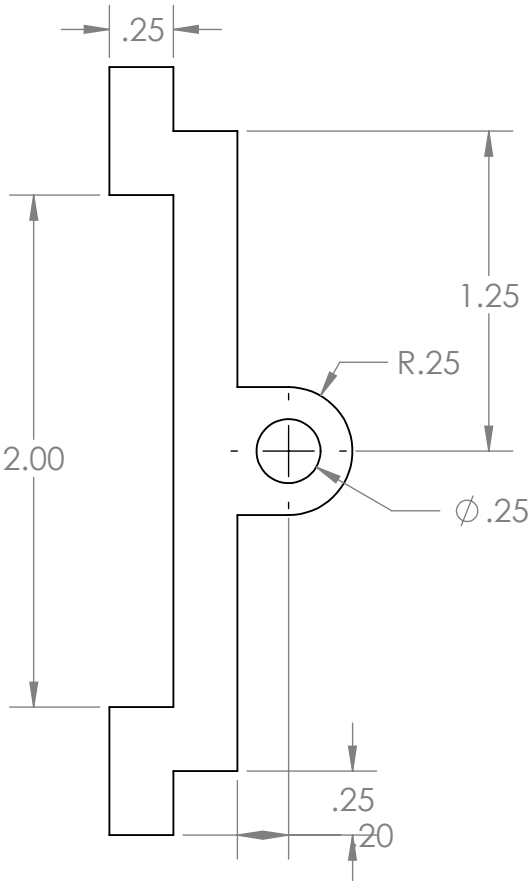
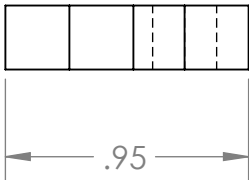
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		ANGULAR: MACH ± BEND ±	ENG APPR.					
		TWO PLACE DECIMAL ± THREE PLACE DECIMAL ±	MFG APPR.					
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.			SIZE DWG. NO. REV B RR602-14 1		
		MATERIAL NYLON	COMMENTS:					
		FINISH						
NEXT ASSY	USED ON					SCALE: 1:0.75 SHEET 18 OF 20		
APPLICATION		DO NOT SCALE DRAWING						

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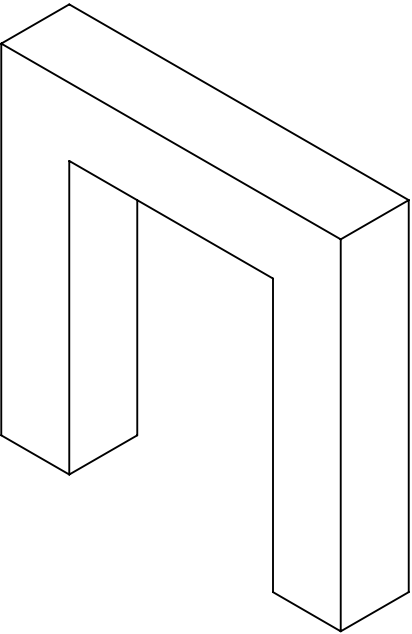
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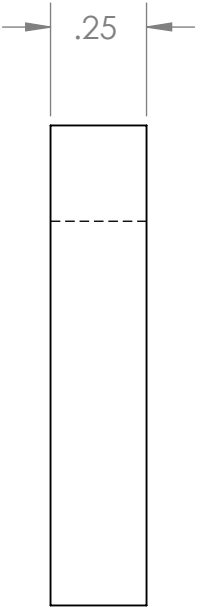
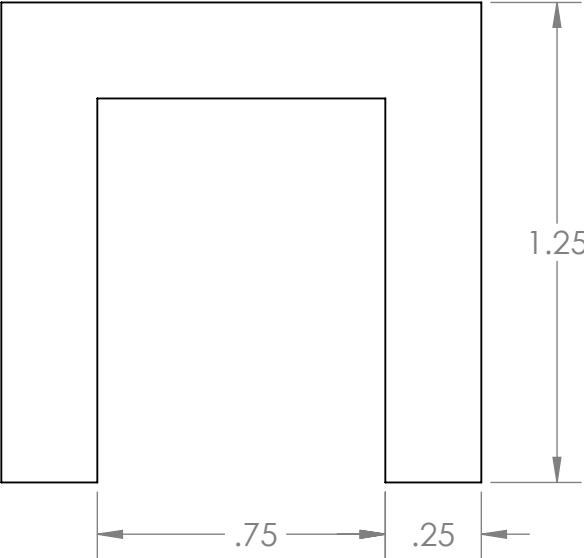
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		FRACTIONAL ±	ENG APPR.					
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		TWO PLACE DECIMAL ±				SIZE DWG. NO. REV B RR602-17 1		
		THREE PLACE DECIMAL ±						
		INTERPRET GEOMETRIC	Q.A.					
		TOLERANCING PER:	COMMENTS:					
		MATERIAL						
		NYLON						
NEXT ASSY	USED ON	FINISH						
APPLICATION		DO NOT SCALE DRAWING				SCALE: 2:1 SHEET 19 OF 20		

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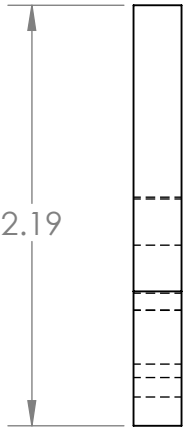
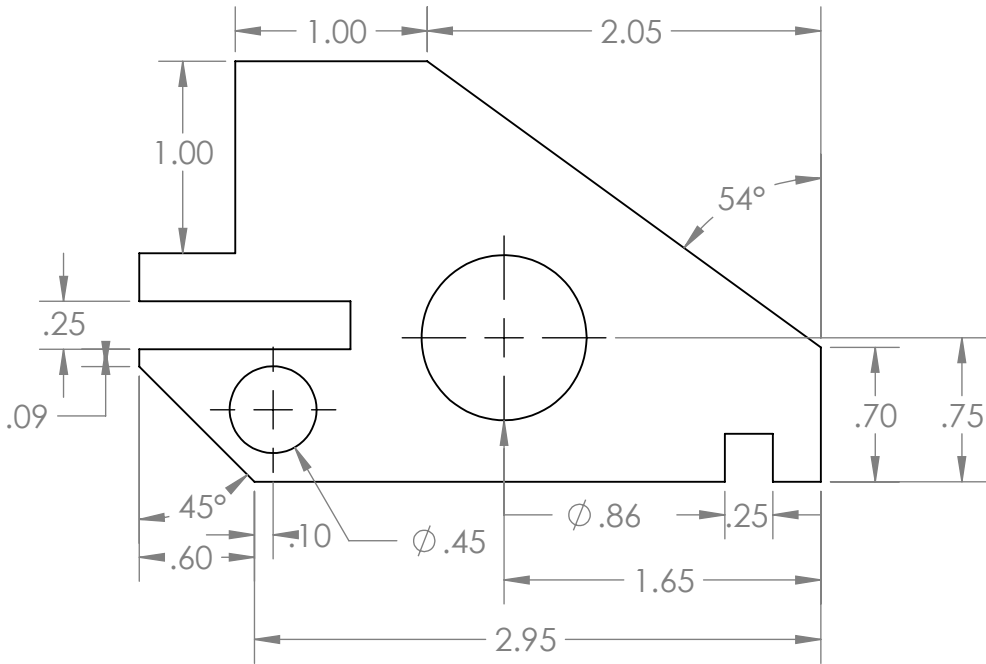
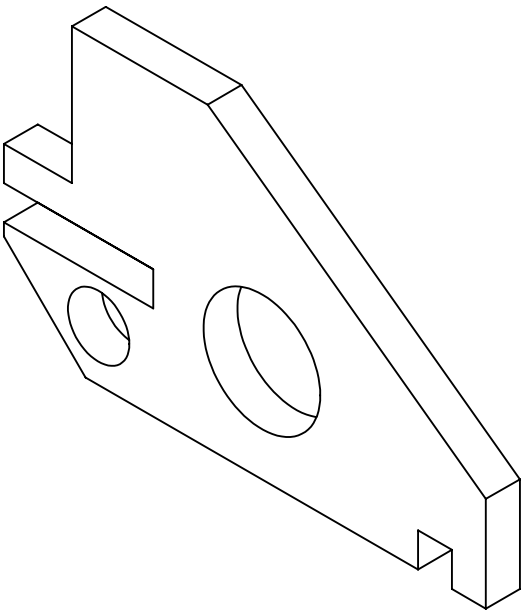
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		INTERPRET GEOMETRIC TOLERANCING PER:	COMMENTS:					
		MATERIAL NYLON						
NEXT ASSY	USED ON	FINISH						
APPLICATION		DO NOT SCALE DRAWING				SCALE: 1:1		SHEET 20 OF 20